SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 3

Section 1: Multiple Choice Questions

1. A linked list contains components, called _____, which are linked to one another.
   a) nodes
   b) arrays
   c) vectors
   d) references

2. The following statement declares p to be _____.
   ```
   int *p;
   ```
   a) an integer
   b) a pointer to an integer
   c) an undefined data type
   d) initialized to NULL

3. The members of a node structure can be accessed with the _____ operator.
   a) [ ]
   b) &
   c) ->
   d) At

4. After execution of the following statements, what value does the variable q contain?
   ```
   p = new int;
   q = p;
   delete p;
   p = NULL;
   ```
   a) the address of p
   b) the address of the deallocated memory cell
   c) the value NULL
   d) q does not point to anything

5. Which of the following will be true when the pointer variable cur references the last node in a linear linked list?
   a) cur == NULL
   b) head == NULL
   c) cur->next == NULL
   d) head->next == NULL

Section 2: Answer ALL questions.

1. Explain the difference between linear and non-linear data structures with examples.
2. Compare and contrast primitive and non-primitive data structures.
3. Define a linked list. Explain the difference between a singly linked list and a doubly linked list.
4. Why is a linked list considered a dynamic data structure? Discuss how this dynamic nature compares to arrays.
5. Discuss the advantages of using a linked list over an array. In what scenarios would you prefer using a linked list over an array?
6. Explain the process of inserting a new node at the beginning, middle, and end of a singly linked list.
7. Linked lists use dynamic memory allocation. Explain how memory is allocated and deallocated in a linked list.